# The Ghost in the Cache: Tinker Tailor LLM Spy

## Forensic Carving of 'Deleted' LLM Chats, Unsubmitted Prompts & Document Blobs from Chromium Storage

**Target Venue:** Black Hat Asia Briefings

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**Official Repository:** https://github.com/sapnadfir-blip/ghost-in-the-cache

# I. The Ephemeral AI Fallacy & LSM-Tree Remanence

• **Cloud Deletion vs Local Persistence:** Clicking 'Delete Chat' updates cloud servers, but Chromium (Chrome, Edge, Brave) and Electron desktop apps retain local telemetry in LevelDB storage.

• **Append-Only LSM Engine:** Writes are appended sequentially to Write-Ahead Logs (.log). 'Deletions' are mere index updates or tombstones; serialized data blocks linger intact until background compaction.

• **Empirical Finding:** 506 text artifacts recovered across 23 profiles; **70.8% (358 artifacts) were already deleted from the web UI**. One artifact persisted for 83 days.

# II. Reverse Engineering V8 Serialization & Framing

• **Varint Decoding:** Protocol Buffer style 7-bit variable-length integer parsing for offsets and string lengths.

• **Tag Identification:** Parsing OneByteString (0x22 ASCII/Latin-1) and TwoByteString (0x63 UTF-16LE byte-length tagged strings).

• **Smi-Shifted Array Keys:** Small Integer keys encoded via left-shift (encoded = actual << 1). Odd actual indices = User Prompts; Even actual indices = Assistant Replies.

• **Nesting Depth Tracking:** Dynamic tracking of object/array depth (0x6f/0x61) prevents premature message truncation on inner end tags (0x7b).

# III. New Endpoint Vulnerability Discoveries

• **Claude Pre-Submission Keystroke Caching:** TipTap editor state serializes in-flight keystrokes to disk in real-time *before* the user clicks 'Send'. 148 draft fragments recovered.

• **Plaintext Document Blobs (.indexeddb.blob/):** Uploaded 10MB PDF documents and proprietary code files stored completely unencrypted, surviving conversation deletion indefinitely.

• **ECDSA Private Keys in ChatGPT Desktop:** com.openai.atlas stores raw WebRTC ECDSA private keys (-----BEGIN PRIVATE KEY-----) in LevelDB logs alongside chat transcripts.

• **Google Gemini Extension (glic):** Stores conversation keys (BARD_EMBED_CHAT_STORAGE_KEY) in Local Storage LevelDB.

# IV. Live Carving Engine & Shadow AI DLP

• **Lock-Free Live Carving:** Read-only binary stream parsing bypasses exclusive LevelDB process locks without disrupting active browser sessions.

• **Real-Time Shadow AI DLP:** On-the-fly extraction and regex classification of leaked AWS access keys, OpenAI API tokens, Slack tokens, JWTs, and PII.

• **HMAC-SHA256 Chain of Custody:** Canonical evidence payload signing, verified client-side using Web Cryptography API (window.crypto.subtle.verify).

• **Sub-163ms Performance:** Full multi-profile scan completes in 147–163 ms, executing before endpoint monitor latency windows.

# V. Defensive Mitigations & Tool Release

• **OS Storage Encryption:** Enforce full-disk encryption (FileVault / BitLocker) and extend DPAPI protections to IndexedDB storage directories.

• **Session Purge Policies:** Configure enterprise GPO/MDM browser policies to purge IndexedDB cache and blob files on browser termination.

• **Endpoint Telemetry Rules:** Deploy EDR behavioral rules monitoring unprivileged access to Chromium IndexedDB directories.

• **Complete Open-Source Release:**
**https://github.com/sapnadfir-blip/ghost-in-the-cache**